

THILAK KUSAMPUDI

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LinkedIn

GitHub

OBJECTIVE

Enthusiastic and highly motivated student with a passion for building real-world projects and solving technical problems. Experienced in developing applications, working with data, and collaborating in team-based environments. Seeking opportunities to apply my skills and contribute to innovative projects.

EDUCATION

Mahindra University

B.Tech in Computer Science and Engineering — CGPA: 6.62/10 (till 5th sem).

Telangana, India

2023 – 2027

Sri Chaitanya Junior College

Telangana, India

2021 – 2023

Intermediate — 96.1%.

Telangana, India

2020 – 2021

Sri Chaitanya Techno School

10th Grade - 100%.

TECHNICAL SKILLS

Programming Languages:	C, Java, JavaScript, Python, HTML, CSS
Frameworks/Libraries:	Django, NumPy, Pandas, Matplotlib, Scikit-learn, Keras
Databases:	MySQL, SQLite
Developer Tools:	Google Colab, VS Code, Power BI
Academic Coursework:	Data Structures, Object-Oriented Programming

PROJECTS

Hybrid Predictive Maintenance System using NASA CMAPSS 2025

- Built a multi-stage predictive maintenance pipeline using K-Means, Logistic Regression, and Random Forest for time-to-failure prediction.
- Engineered a normalized risk score combining classification probabilities and regression outputs.
- Performed feature scaling, imputation, PCA visualization, and model evaluation using RMSE, MAE, and R^2 .

Power Demand Forecasting & Anomaly Detection Dashboard (OPSD) 2025

- Developed time-series forecasting using SARIMA and GRU for 24-hour ahead electricity demand prediction.
- Implemented anomaly detection using z-score residual analysis and confidence interval deviation.
- Built an interactive Streamlit dashboard with KPIs, forecasts, and downloadable reports.

Driver Drowsiness Detection 2025

- Built a CNN-based model using TensorFlow and OpenCV to detect driver drowsiness from live camera feeds.
- Integrated an alerting module with real-time audio notifications to prevent fatigue-related accidents.
- Optimized inference for edge devices, improving frame-rate consistency and overall responsiveness.

Multi-Round Interview Agent 2026

- Built an AI-driven interview automation platform using Flask and OpenAI GPT, supporting multi-round assessments and AI-based code evaluation, reducing HR screening time by 80%.
- Implemented secure RESTful APIs with session-based authentication and role-based access control, ensuring scalable, consistent, and secure candidate evaluations across different hiring roles.
- Designed a responsive ChatGPT-like conversational interface and admin dashboard featuring candidate management, real-time analytics, and performance insights for streamlined hiring workflows.

CERTIFICATIONS

- AI Hackathon organized by Forge Alumnus, held at Microsoft, Hyderabad
- Data Analytics Essentials - Cisco Networking Academy
- Ethical Hacking Workshop at BITS Hyderabad